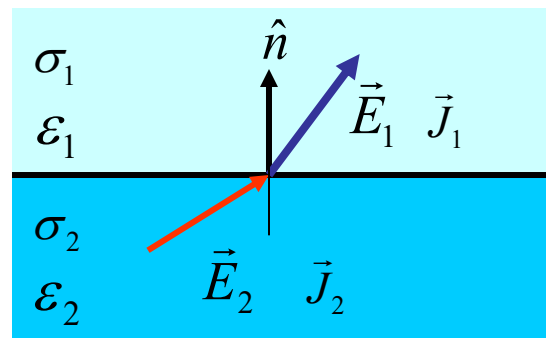


3.3 恒定电流场的边界条件

$$\oiint_S \vec{J} \cdot d\vec{S} = 0 \Rightarrow J_{1n} = J_{2n}$$

$$\oint_l \vec{E} \cdot d\vec{l} = 0 \Rightarrow E_{1t} = E_{2t}$$



两种典型的边界

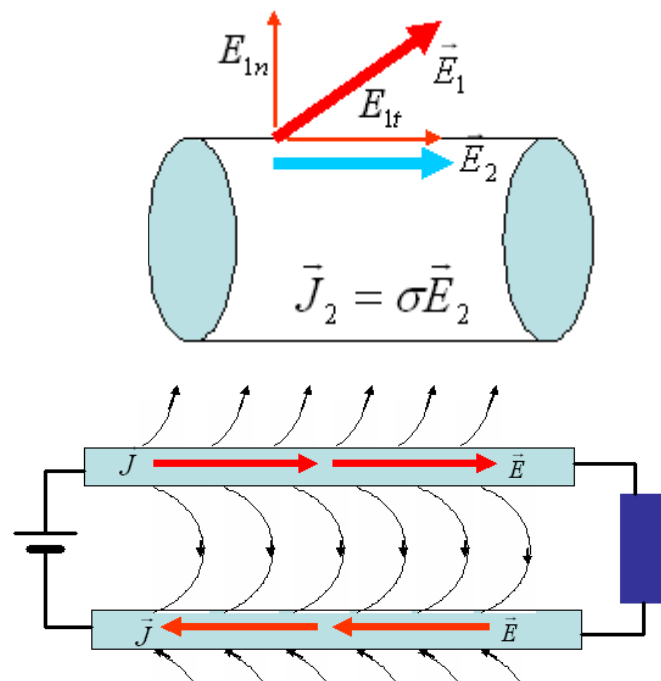
1) 理想介质-导体边界

$$\sigma_1 = 0$$

$$J_1 = 0 \quad J_{1n} = J_{2n} \quad J_{2n} = 0 \quad E_{2n} = 0$$

$$\vec{J}_2 = J_{2t} \hat{t} \quad \vec{E}_2 = E_{2t} \hat{t}$$

$$\rho_s = D_{1n}$$



2) 一般导体-理想导体边界

$$\sigma_2 = \infty \quad \vec{J}_2 = \sigma_2 \vec{E}_2 \quad \vec{E}_2 = 0$$

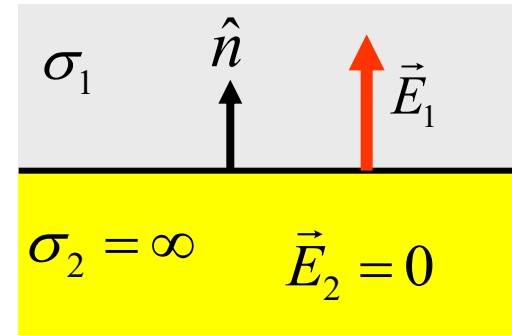
$$E_{1t} = E_{2t} = 0$$

$$\vec{E}_1 = E_{1n} \hat{n}$$

电位的边界条件

$$\sigma_1 E_{1n} = \sigma_2 E_{2n} \quad \Rightarrow \quad \sigma_1 \frac{\partial \varphi_1}{\partial n} = \sigma_2 \frac{\partial \varphi_2}{\partial n}$$

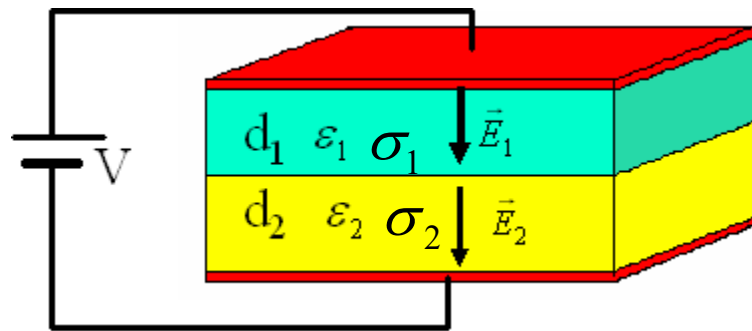
$$E_{1t} = E_{2t} \quad \Rightarrow \quad \varphi_1 = \varphi_2$$



例1. 已知平板电容器两导电平板面积为A，之间填充两层非理想介质，厚度分别为 d_1 和 d_2 与电导率分别为 σ_1 和 σ_2 。当两导电平板之间加电压V时，求电场强度及电容器的漏电导。

解： 解法 $G = I / V$

设导电平板之间的两层媒质中电场均为匀强场，那么由边界条件及电压关系得



$$\sigma_1 E_1 = \sigma_2 E_2$$

$$E_1 d_1 + E_2 d_2 = V$$

$$E_1 = \frac{\sigma_2}{d_1 \sigma_2 + d_2 \sigma_1} V \quad E_2 = \frac{\sigma_1}{d_1 \sigma_2 + d_2 \sigma_1} V$$

$$J_1 = \sigma_1 E_1 = \frac{\sigma_1 \sigma_2}{d_1 \sigma_2 + d_2 \sigma_1} V = J_2$$

$$I = JA = \frac{\sigma_1 \sigma_2}{d_1 \sigma_2 + d_2 \sigma_1} VA \quad G = \frac{I}{V} = \frac{\sigma_1 \sigma_2 A}{d_1 \sigma_2 + d_2 \sigma_1}$$

3.4 能量损耗与电动势

1) 焦耳定律

导体电流场中自由电荷在电场的作用下，
电场力所作的功：

$$dA = dq \vec{E} \cdot d\vec{l}$$

电场提供的功率为

$$P = \frac{dA}{dt} = \frac{dq}{dt} \vec{E} \cdot d\vec{l} = I \vec{E} \cdot d\vec{l} = (\vec{J} \cdot d\vec{S}) \cdot \vec{E} \cdot d\vec{l} = \vec{J} \cdot \vec{E} \cdot dV$$

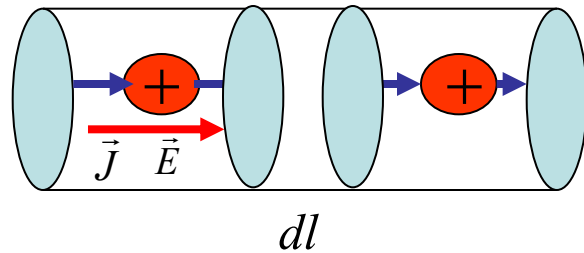
功率密度：电场为单位体积中的电荷提供的功率：
$$p = \vec{J} \cdot \vec{E}$$

功率密度等于电场强度与电流体密度的标积

电场对电荷做了功，电场能量减少，减少的能量转化为热能。在导电媒质中电场提供的功率密度 p 就等于在单位时间单位体积中产生的热能，也就是**损耗功率密度**。

$$p = \vec{E} \cdot \vec{J} = \sigma E^2 = \frac{J^2}{\sigma}$$

焦耳定律的微分形式



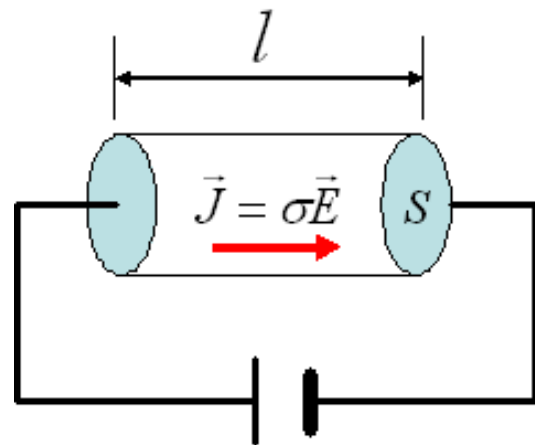
损耗功率密度: $p = \vec{E} \cdot \vec{J} = \sigma E^2 = \frac{J^2}{\sigma}$

体积 V 中导电媒质总功率损耗为 $P = \int_V \sigma E^2 dV$

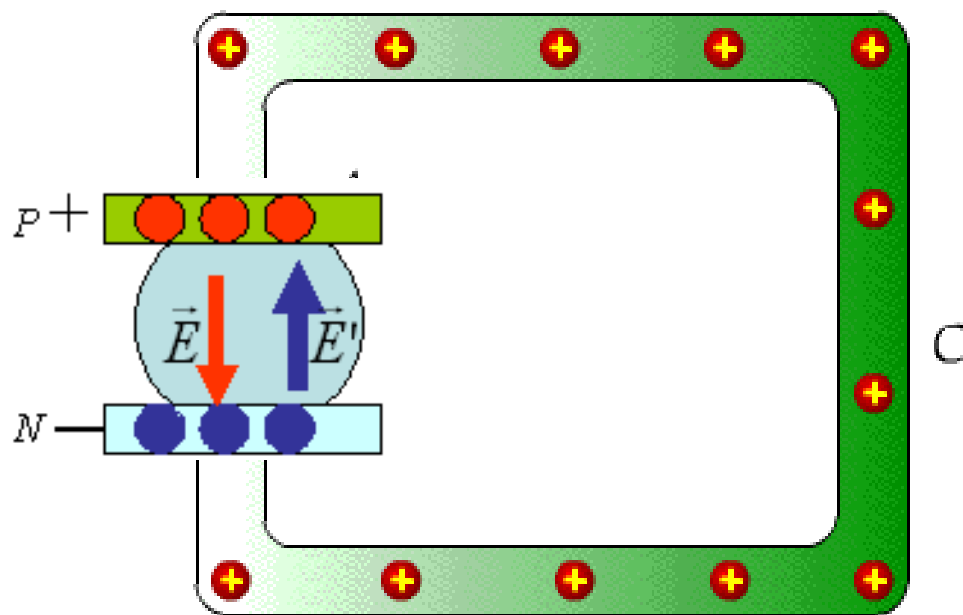
对于长度为 L 、截面积为 S 的均匀导体, 如果在导体两端加电压 U , 导体中为匀强电场, 电流分布均匀, 功率损耗为:

$$P = \int_V \sigma E^2 dV = \sigma E^2 Sl = JSEl = IU \\ = I^2 R$$

电路中的焦耳定律



2) 电动势



为了维持电流流动，在正负极板之间放置外源，外源中存在对电荷有作用力的**非静电力**，这种外源非静电力做功反抗极板之间的静电场力，将通过导线从正极板移到负极板的正电荷再经过电源内部又移回到正极板，保持极板上的电荷密度不变，从而也就使导电媒质中的电场和电流维持恒定不变。

外源中的非静电力表现为对电荷的作用力，可以认为这种非静电力是由外源中存在的外电场产生的。

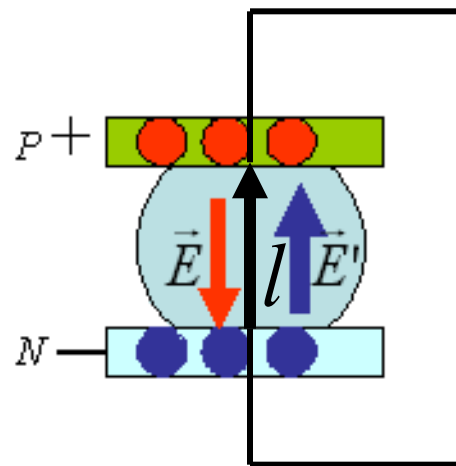
在电源中，外电场与极板之间电荷产生的电场刚好相反

$$\vec{E}' = -\vec{E}$$

电源的电动势： 电源将单位正电荷从电源的负极 N 移到正极 P 所做的功

$$\varepsilon = \int_N^P \vec{E}' \cdot d\vec{l}$$

$$= \int_P^N \vec{E} \cdot d\vec{l} = U$$



3.5 恒定电流场与静电场的比拟

均匀导电媒质中的恒定
电流场方程

$$\oint_S \vec{J} \cdot d\vec{S} = 0$$

$$\oint_l \vec{E} \cdot d\vec{l} = 0$$

$$\nabla \cdot \vec{J} = 0$$

$$\nabla \times \vec{E} = 0$$

$$\vec{J} = \sigma \vec{E}$$

$$J_{1n} = J_{2n}$$

$$E_{1t} = E_{2t}$$

无源区均匀介质中的静电场方程

$$\oint_S \vec{D} \cdot d\vec{S} = 0$$

$$\oint_l \vec{E} \cdot d\vec{l} = 0$$

$$\nabla \cdot \vec{D} = 0$$

$$\nabla \times \vec{E} = 0$$

$$\vec{D} = \varepsilon \vec{E}$$

$$D_{1n} = D_{2n}$$

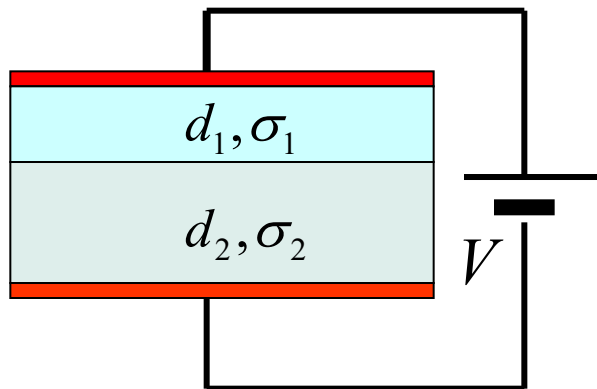
$$E_{1t} = E_{2t}$$

$$\vec{J} \leftrightarrow \vec{D}$$

$$\vec{E} \leftrightarrow \vec{E}$$

$$\sigma \leftrightarrow \varepsilon$$

例1



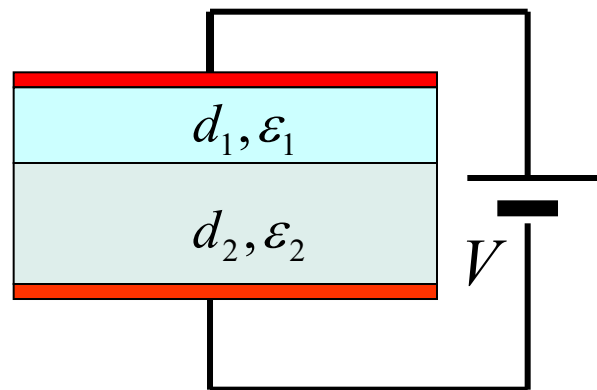
$$E_1 = \frac{\sigma_2}{d_1\sigma_2 + d_2\sigma_1} V$$

$$E_2 = \frac{\sigma_1}{d_1\sigma_2 + d_2\sigma_1} V$$

$$J_1 = \sigma_1 E_1 = \frac{\sigma_1\sigma_2}{d_1\sigma_2 + d_2\sigma_1} V$$

$$I = JA = \frac{\sigma_1\sigma_2}{d_1\sigma_2 + d_2\sigma_1} VA$$

$$G = \frac{I}{V} = \frac{\sigma_1\sigma_2 A}{d_1\sigma_2 + d_2\sigma_1}$$



$$E_1 = \frac{\epsilon_2}{d_1\epsilon_2 + d_2\epsilon_1} V$$

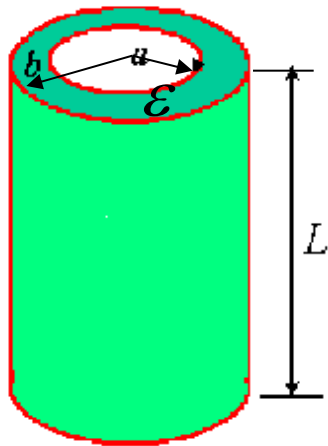
$$E_2 = \frac{\epsilon_1}{d_1\epsilon_2 + d_2\epsilon_1} V$$

$$D_1 = \epsilon_1 E_1 = \frac{\epsilon_1\epsilon_2}{d_1\epsilon_2 + d_2\epsilon_1} V$$

$$q = \rho_s A = D_n A = \frac{\epsilon_1\epsilon_2}{d_1\epsilon_2 + d_2\epsilon_1} VA$$

$$C = \frac{q}{V} = \frac{\epsilon_1\epsilon_2 A}{d_1\epsilon_2 + \epsilon_2\sigma_1}$$

例2 已知电压求E



设 q

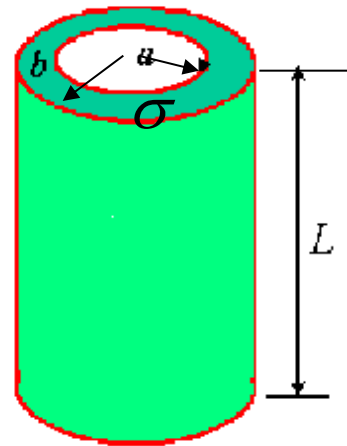
$$\oint_S \vec{D} \cdot d\vec{S} = 2\pi\rho L D_\rho = q$$

$$\vec{D} = \frac{q\hat{\rho}}{2\pi L\rho} \quad \vec{D} = \epsilon\vec{E}$$

$$\vec{E} = \frac{q\hat{\rho}}{2\pi\epsilon L\rho}$$

$$V = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b \frac{q d\rho}{2\pi\epsilon L\rho} = \frac{q}{2\pi\epsilon L} \ln \frac{b}{a}$$

$$\vec{E} = \hat{\rho} \frac{V}{\ln \frac{b}{a}} \frac{1}{\rho} \quad C = \frac{q}{V} = 2\pi\epsilon L \ln \frac{a}{b}$$



设 I

$$\iint_S \vec{J} \cdot d\vec{S} = 2\pi\rho L J_\rho = I$$

$$\vec{J} = \frac{I\hat{\rho}}{2\pi L\rho} \quad \vec{J} = \sigma\vec{E}$$

$$\vec{E} = \frac{I\hat{\rho}}{2\pi\sigma L\rho}$$

$$V = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b \frac{I d\rho}{2\pi\sigma L\rho} = \frac{I}{2\pi\sigma L} \ln \frac{b}{a}$$

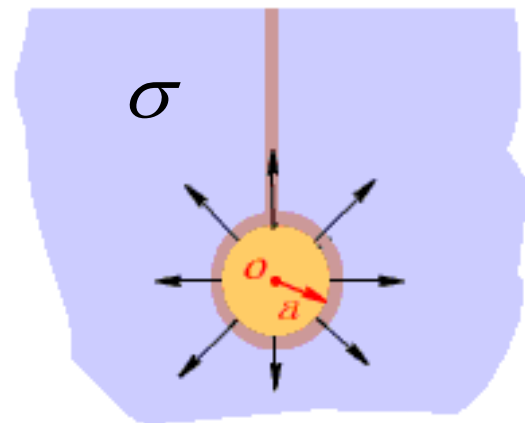
$$\vec{E} = \hat{\rho} \frac{V}{\ln \frac{b}{a}} \frac{1}{\rho} \quad G = \frac{I}{V} = 2\pi\sigma L \ln \frac{a}{b}$$

$$G = \frac{I}{U} = \frac{\int_s \vec{J} \cdot d\vec{S}}{\int_N^P \vec{E} \cdot d\vec{l}} = \frac{\sigma \int_s \vec{E} \cdot d\vec{S}}{\int_N^P \vec{E} \cdot d\vec{l}} \quad C = \frac{q}{U} = \frac{\int_s \vec{D} \cdot d\vec{S}}{\int_N^P \vec{E} \cdot d\vec{l}} = \frac{\varepsilon \int_s \vec{E} \cdot d\vec{S}}{\int_N^P \vec{E} \cdot d\vec{l}}$$

例3 深埋球形接地器的接地电阻

解法1 静电比拟 $C = \frac{q}{\Phi} = 4\pi\varepsilon a$

$$G = \frac{I}{\Phi} \leftrightarrow C = \frac{q}{\Phi} \quad R = \frac{\Phi}{I} = \frac{1}{G} = \frac{1}{4\pi\sigma a}$$



深埋球形接地器

解法2 $R = \frac{\Phi}{I}$

设 I $I = \iint_S \vec{J} \cdot d\vec{S} = J_r 4\pi r^2$ $\vec{J} = \frac{I}{4\pi r^2} \hat{r}$

$$\vec{E} = \frac{\vec{J}}{\sigma} = \frac{I}{4\pi\sigma r^2} \hat{r} \quad \Phi = \int_a^\infty \vec{E} \cdot d\vec{l} = \frac{I}{4\pi\sigma a}$$

$$R = \frac{\Phi}{I} = \frac{1}{4\pi\sigma a}$$

总结

恒定电流场方程

恒定电流场边界条件

损耗功率密度和电动势

恒定电流场与静电场比拟